

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250277588

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

Ge; Dick et al.

COOKING APPLIANCE WITH AN AUXILIARY HEATING ELEMENT

Abstract

A cooking appliance (10) includes a body (12) that defines a cooking cavity (14), a first electrical connection feature (16) coupled to the body (12), an auxiliary heating element (18), and a second electrical connection feature (20) coupled to the auxiliary heating element (18). The auxiliary heating element (18) is operable between a use position and a removed position. In the use position, the auxiliary heating element (18) is positioned within the cooking cavity (14). In the removed position, the auxiliary heating element (18) is positioned outside of the cooking cavity (14). The second electrical connection feature (20) is operable to move relative to the auxiliary heating element (18) between engaged and disengaged positions. Contact between the first electrical connection feature (16) and the second electrical connection feature (20) as the auxiliary heating element (18) enters the use position causes the second electrical connection feature (20) to enter the engaged position.

Inventors:	Ge; Dick (Shenzhen, CN), Huang; Benny (Shenzhen, CN), Li; Ryan (Shenzhen, CN)
Applicant:	WHIRLPOOL CORPORATION (BENTON HARBOR, MI)
Family ID:	88646167
Assignee:	WHIRLPOOL CORPORATION (BENTON HARBOR, MI)
Appl. No.:	18/858899
Filed (or PCT Filed):	May 06, 2022
PCT No.:	PCT/CN2022/091167

Publication Classification

Int. Cl.: F24C7/06 (20060101); F24C15/00 (20060101)

U.S. Cl.:

CPC F24C7/06 (20130101); F24C15/007 (20130101);

Background/Summary

BACKGROUND OF THE DISCLOSURE

[0001] The present disclosure generally relates to a cooking appliance and, more specifically, to a cooking appliance that includes an auxiliary heating element.

SUMMARY OF THE DISCLOSURE

[0002] According to one aspect of the present disclosure, a cooking appliance includes a body that defines a cooking cavity, a first electrical connection feature coupled to the body, an auxiliary heating element, and a second electrical connection feature coupled to the auxiliary heating element. The auxiliary heating element is operable between a use position and a removed position. In the use position, the auxiliary heating element is positioned within the cooking cavity. In the removed position, the auxiliary heating element is positioned outside of the cooking cavity. The second electrical connection feature is operable to move relative to the auxiliary heating element between engaged and disengaged positions. Contact between the first electrical connection feature and the second electrical connection feature as the auxiliary heating element enters the use position causes the second electrical connection feature to enter the engaged position.

[0003] According to another aspect of the present disclosure, a cooking appliance includes a body that defines a cooking cavity, a first electrical connection feature coupled to the body, an auxiliary heating element, and a second electrical connection feature coupled to the auxiliary heating element. The auxiliary heating element is operable between a use position and a removed position. In the use position, the auxiliary heating element is positioned within the cooking cavity. In the removed position, the auxiliary heating element is positioned outside of the cooking cavity. The second electrical connection feature is operable to move relative to the auxiliary heating element between engaged and disengaged positions and biased toward the disengaged position. Contact between the first electrical connection feature and the second electrical connection feature as the auxiliary heating element enters the use position causes the second electrical connection feature to enter the engaged position. In the use position of the auxiliary heating element, a support feature of the body generally prevents forward movement of the auxiliary heating element, such that the auxiliary heating element is retained in the engaged position.

[0004] According to yet another aspect of the present disclosure, a cooking appliance includes a body that defines a cooking cavity and includes a support feature in communication with the cooking cavity, and an auxiliary heating element operable between a removed position and a use position. In the removed position, the auxiliary heating element is positioned outside of the cooking cavity. In the use position, the auxiliary heating element is positioned within the cooking cavity in contact with the support feature, such that forward movement of the auxiliary heating element out of the use position is generally prevented.

[0005] These and other features, advantages, and objects of the present disclosure will be further understood and appreciated by those skilled in the art by reference to the following specification, claims, and appended drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] In the drawings:

[0007] FIG. 1 is a front elevational view of a cooking appliance that includes an auxiliary heating element in a use position;

[0008] FIG. 2 is a front elevational view of a cooking appliance that includes an auxiliary heating element in a removed position;

[0009] FIG. 3 is a cross-sectional plan view of an electrical connection feature of an auxiliary heating element of a cooking appliance in a disengaged position;

[0010] FIG. 4 is a cross-sectional plan view of an electrical connection feature of an auxiliary heating element of a cooking appliance in an engaged position, abutting a corresponding electrical connection feature of the cooking appliance; and

[0011] FIG. 5 is a cross-sectional view of a portion of the cooking appliance of FIG. 1 taken at line V-V and illustrating the auxiliary heating element in the use position.

[0012] The components in the figures are not necessarily to scale, emphasis instead being placed upon illustrating the principles described herein.

DETAILED DESCRIPTION

[0013] The present illustrated embodiments reside primarily in combinations apparatus components related to a cooking appliance with an auxiliary heating element. Accordingly, the apparatus components have been represented, where appropriate, by conventional symbols in the drawings, showing only those specific details that are pertinent to understanding the embodiments of the present disclosure so as not to obscure the disclosure with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein. Further, like numerals in the description and drawings represent like elements.

[0014] For purposes of description herein, the terms “upper,” “lower,” “right,” “left,” “rear,” “front,” “vertical,” “horizontal,” and derivatives thereof shall relate to the disclosure as oriented in FIG. 1. Unless stated otherwise, the term “front” shall refer to the surface of the element closer to an intended viewer, and the term “rear” shall refer to the surface of the element further from the intended viewer. Further, the term “forward movement” and derivatives thereof shall refer to movement toward the intended viewer and perpendicular to the vertical direction and the left-right horizontal direction based on the disclosure as oriented in FIG. 1. The term “rearward movement” and derivatives thereof shall refer to movement away from the intended viewer and perpendicular to the vertical direction and the left-right horizontal direction based on the disclosure as oriented in FIG. 1. The term “downward movement” and derivatives thereof shall refer to movement from upper to lower in a direction perpendicular to the forward movement direction and the left-right horizontal direction based on the disclosure as oriented in FIG. 1. However, it is to be understood that the disclosure may assume various alternative orientations, except where expressly specified to the contrary. It is also to be understood that the specific devices and processes illustrated in the attached drawings, and described in the following specification, are simply exemplary embodiments of the inventive concepts defined in the appended claims. Hence, specific dimensions and other physical characteristics relating to the embodiments disclosed herein are not to be considered as limiting, unless the claims expressly state otherwise.

[0015] The terms “including,” “comprises,” “comprising,” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. An element preceded by “comprises a . . .” does not, without more constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises the element.

[0016] Referring now to FIGS. 1-5, reference numeral **10** generally designates a cooking appliance. The cooking appliance **10** includes a body **12** that defines a cooking cavity **14**. A first electrical

connection feature **16** is coupled to the body **12**. An auxiliary heating element **18** is operable between a use position, wherein the auxiliary heating element **18** is positioned within the cooking cavity **14**, and a removed position, wherein the auxiliary heating element **18** is positioned outside of the cooking cavity **14**. A second electrical connection feature **20** is coupled to the auxiliary heating element **18**. The second electrical connection feature **20** is operable to move relative to the auxiliary heating element **18** between an engaged position and a disengaged position. Contact between the first electrical connection feature **16** and the second electrical connection feature **20** as the auxiliary heating element **18** enters the use position causes the second electrical connection feature **20** to enter the engaged position.

[0017] Referring now to FIGS. **1** and **2**, the cooking appliance **10** includes the body **12**. As illustrated in FIGS. **1** and **2**, the body **12** defines a cooking cavity **14**. The cooking cavity **14** is accessible via an opening **22** defined by the body **12** proximate to a front **24** of the cooking appliance **10**. In various implementations, the cooking appliance **10** can include a door (not shown) that is configured to selectively cover the opening **22**. The cooking cavity **14** is configured to receive food therein for cooking. The cooking appliance **10** can include at least one heating element **26** that is configured to produce heat for heating the cooking cavity **14**. In various implementations, the heating element **26** may be an electrical heating element that is powered by electricity. A variety of types of heating elements **26** are contemplated. As illustrated in FIGS. **1** and **2**, the heating element **26** is fixedly positioned within the cooking cavity **14** proximate to an interior bottom wall **28** of the body **12**.

[0018] As further illustrated in FIGS. **1** and **2**, the body **12** can include the interior bottom wall **28**, an interior top wall **30** opposite the interior bottom wall **28**, two interior side walls **32** that extend between the interior bottom and top walls **28**, **30**, and an interior rear wall **34** that is positioned opposite the opening **22** to the cooking cavity **14**. The interior bottom, top, rear, and side walls **28**, **30**, **34**, **32** define the cooking cavity **14**, as illustrated in FIGS. **1** and **2**. In various embodiments, the body **12** of the cooking appliance **10** is an assembly of a plurality of components.

[0019] Referring now to FIGS. **1**, **2**, and **5**, the body **12** of the cooking appliance **10** includes a support feature **36**. The support feature **36** is in communication with the cooking cavity **14**. As described further herein, the support feature **36** is configured to support the auxiliary heating element **18** in the use position of the auxiliary heating element **18**. In various implementations, the support feature **36** can be positioned on the interior side wall **32** of the body **12** of the cooking appliance **10**. As illustrated in FIGS. **1**, **2**, and **5**, the body **12** of the cooking appliance **10** can include a plurality of support features **36**, in some implementations. As illustrated in FIGS. **1** and **2**, both interior side walls **32** of the body **12** include respective support features **36**. Further, as illustrated in FIG. **4**, the interior side wall **32** includes two support features **36** that are vertically aligned and in a spaced relationship with each other. In the illustrated embodiment, the support feature **36** extends outward from the adjacent portion of the interior side wall **32** into the cooking cavity **14**.

[0020] In the embodiment illustrated in FIG. **5**, the support feature **36** is positioned on the interior side wall **32** of the body **12** and includes a first shelf portion **38**, a second shelf portion **40**, and a saddle portion **42**. The second shelf portion **40** is positioned between the first shelf portion **38** and the interior rear wall **34** of the body **12**. The saddle portion **42** extends between the first shelf portion **38** and the second shelf portion **40**. As illustrated in FIG. **5**, the saddle portion **42** extends vertically downward and toward the interior side wall **32** that the support feature **36** is positioned on from both the first shelf portion **38** and the second shelf portion **40**.

[0021] Referring still to FIG. **5**, in various implementations, the body **12** of the cooking appliance **10** can include a protrusion **44**. In various embodiments, the body **12** of the cooking appliance **10** includes a plurality of protrusions **44**. The protrusion **44** of the body **12** is in communication with the cooking cavity **14** and extends outward from at least one of the interior rear wall **34** and the interior side wall **32** of the body **12**. For example, in the embodiment illustrated in FIG. **5**, the body

12 includes a plurality of protrusions **44** that extend outward from the interior side wall **32** into the cooking cavity **14**. As illustrated, each of the protrusions **44** are positioned generally between the interior rear wall **34** and the support feature **36**. In other words, the protrusions **44** are positioned rearward of the support feature **36**. As described further herein, the protrusions **44** may cooperate with the support feature **36** to guide movement of the auxiliary heating element **18** into and out of the use position, and support the auxiliary heating element **18** in the use position.

[0022] Referring now to FIGS. 1-5, the cooking appliance **10** includes the first electrical connection feature **16**. The first electrical connection feature **16** is coupled to the body **12** of the cooking appliance **10**. In various implementations, the first electrical connection feature **16** is in communication with the cooking cavity **14** defined by the body **12** of the cooking appliance **10**. For example, as illustrated in FIG. 2, the first electrical connection feature **16** is disposed on the interior rear wall **34** of the body **12**. It is contemplated that the first electrical connection feature **16** may be positioned on various portions of the body **12** that are in communication with the cooking cavity **14** (e.g., interior side wall **32**, etc.). As described further herein, the first electrical connection feature **16** may be operable to contact the second electrical connection feature **20** to provide electricity to the auxiliary heating element **18**.

[0023] Referring now to FIGS. 1 and 2, the cooking appliance **10** includes the auxiliary heating element **18**. The auxiliary heating element **18** may be selectively coupled to the body **12** within the cooking cavity **14**. In other words, the auxiliary heating element **18** may be intended to be removed from the cooking cavity **14**. As illustrated in FIGS. 1 and 2, the auxiliary heating element **18** is operable between the use position (FIG. 1), wherein the auxiliary heating element **18** is positioned within the cooking cavity **14**, and the removed position (FIG. 2), wherein the auxiliary heating element **18** is positioned outside of the cooking cavity **14**.

[0024] Referring now to FIGS. 1-5, the second electrical connection feature **20** is coupled to the auxiliary heating element **18**. The second electrical connection feature **20** is configured to contact the first electrical connection feature **16** in the use position of the auxiliary heating element **18**. In some implementations, the second electrical connection feature **20** is operable to move relative to the auxiliary heating element **18** between the engaged position and the disengaged position. Contact between the first electrical connection feature **16** and the second electrical connection feature **20**, as the auxiliary heating element **18** enters the use position, can cause the second electrical connection feature **20** to enter the engaged position. In some implementations, the second electrical connection feature **20** is biased toward the disengaged position. In some embodiments, the second electrical connection feature **20** moves toward the auxiliary heating element **18** from the disengaged position to the engaged position.

[0025] A variety of types of electrical connection features are contemplated. For example, the second electrical connection feature **20** can be magnetic and mutually correspond with a magnetic first electrical connection feature **16**. In the embodiment illustrated in FIGS. 3 and 4, the second electrical connection feature **20** is a pogo pin **46**. The pogo pin **46** is coupled to the auxiliary heating element **18** and is operable to be depressed from the disengaged position, as illustrated in FIG. 3, to the engaged position, as illustrated in FIG. 4, due to contact between the pogo pin **46** and the first electrical connection feature **16** as the auxiliary heating element **18** is moved from the removed position to the use position. In the engaged position of the second electrical connection feature **20**, electricity is supplied to the auxiliary heating element **18** via the connection between the first and second electrical connection features **16**, **20**. As illustrated in FIGS. 3 and 4, the pogo pin **46** moves toward the auxiliary heating element **18** from the disengaged position to the engaged position.

[0026] Referring now to FIGS. 1, 2, and 5, the auxiliary heating element **18** can include a panel portion **48** and a stop feature **50** that is coupled to the panel portion **48**. As illustrated in FIG. 2, the stop feature **50** may extend generally downward from the panel portion **48** and may be positioned proximate to a lateral side **52** of the panel portion **48**. In various implementations, the panel portion

48 and/or the stop feature **50** can be configured to contact the support feature **36** and/or one or more protrusions **44**, as described further herein.

[0027] Referring now to FIGS. **1** and **5**, in various implementations, the support feature **36** of the body **12** is configured to support the auxiliary heating element **18** in the use position of the auxiliary heating element **18**. In some embodiments, the support feature **36** contacts the auxiliary heating element **18** in the use position of the auxiliary heating element **18**, such that the support feature **36** generally prevents forward movement of the auxiliary heating element **18** out of the use position. In some implementations, the support feature **36** contacts the auxiliary heating element **18** in the use position of the auxiliary heating element **18**, such that the support feature **36** generally prevents downward movement of the auxiliary heating element **18** within the cooking cavity **14**.

[0028] Referring now to FIGS. **1**, **2**, and **5**, in some implementations, wherein the auxiliary heating element **18** includes the panel portion **48** and the stop feature **50** coupled to the panel portion **48**, contact between the support feature **36** and the stop feature **50** of the auxiliary heating element **18** can generally prevent forward movement of the auxiliary heating element **18** out of the use position. As illustrated in FIG. **5**, the support feature **36** is positioned on the interior side wall **32** and includes the first shelf portion **38**, the second shelf portion **40**, and the saddle portion **42** extending between the first and second shelf portions **38**, **40**. In the illustrated embodiment, the stop feature **50** of the auxiliary heating element **18** contacts the saddle portion **42** proximate to the first shelf portion **38** in the use position of the auxiliary heating element **18**, such that the saddle portion **42** generally prevents forward movement of the auxiliary heating element **18** out of the use position. As further illustrated in FIG. **5**, the panel portion **48** of the auxiliary heating element **18** rests upon the first and second shelf portions **38**, **40**, such that the first and second shelf portions **38**, **40** generally prevent downward movement of the auxiliary heating element **18** within the cooking cavity **14** in the use position of the auxiliary heating element **18**.

[0029] In the embodiment illustrated in FIG. **5**, the second electrical connection feature **20** includes the pogo pin **46**, which is contacting the first electrical connection feature **16** in the engaged position with the auxiliary heating element **18** in the use position. The pogo pin **46** is biased toward the interior rear wall **34** of the body **12**. Accordingly, the support feature **36** generally preventing forward movement of the auxiliary heating element **18** out of the use position generally prevents the pogo pin **46** from moving with its biasing force from the engaged position to the disengaged position. It is contemplated that the second electrical connection feature **20** may be biased toward the disengaged position in a direction that is opposite the direction that the support feature **36** generally prevents the auxiliary heating element **18** from moving out of the use position. Various directions are contemplated.

[0030] As illustrated in FIG. **5**, the auxiliary heating element **18** is operable to pivot upward to a position in which the support feature **36** does not prevent forward movement of the auxiliary heating element **18**. The protrusion **44** nearest to the interior rear wall **34** of the body **12** acts as a fulcrum for the pivotal movement of the auxiliary heating element **18** (shown in phantom).

[0031] In operation of an exemplary embodiment of the cooking appliance **10**, a user, desiring to supplement the permanent heating element **26** within the cooking cavity **14** of the cooking appliance **10**, moves the auxiliary heating element **18** from the removed position to the use position. In the use position, the second electrical connection feature **20** coupled to the auxiliary heating element **18** contacts the first electrical connection feature **16**, such that the second electrical connection feature **20** is maintained in the engaged position. The support feature **36** of the body **12** of the cooking appliance **10** supports the auxiliary heating element **18** and generally prevents forward movement of the auxiliary heating element **18** out of the use position. Upon desiring to remove the auxiliary heating element **18** from the use position, the user tilts the forward end of the auxiliary heating element **18** upward, using the protrusion **44** extending outward from the interior side wall **32** of the body **12** as a fulcrum until the support feature **36** no longer prevents forward movement of the auxiliary heating element **18** out of the use position. The user then slides the

auxiliary heating element **18** out of the cooking cavity **14** to the removed position.

[0032] According to an aspect of the present disclosure, a cooking appliance includes a body that defines a cooking cavity, a first electrical connection feature coupled to the body, an auxiliary heating element, and a second electrical connection feature coupled to the auxiliary heating element. The auxiliary heating element is operable between a use position and a removed position. In the use position, the auxiliary heating element is positioned within the cooking cavity. In the removed position, the auxiliary heating element is positioned outside of the cooking cavity. The second electrical connection feature is operable to move relative to the auxiliary heating element between engaged and disengaged positions. Contact between the first electrical connection feature and the second electrical connection feature as the auxiliary heating element enters the use position causes the second electrical connection feature to enter the engaged position.

[0033] According to another aspect of the present disclosure, the second electrical connection feature is biased toward the disengaged position.

[0034] According to another aspect of the present disclosure, the second electrical connection feature moves toward the auxiliary heating element from the disengaged position to the engaged position.

[0035] According to another aspect of the present disclosure, the second electrical connection feature includes a pogo pin.

[0036] According to another aspect of the present disclosure, the body includes a support feature in communication with the cooking cavity and configured to support the auxiliary heating element in the use position of the auxiliary heating element.

[0037] According to another aspect of the present disclosure, the support feature contacts the auxiliary heating element in the use position of the auxiliary heating element, such that the support feature generally prevents forward movement of the auxiliary heating element out of the use position.

[0038] According to another aspect of the present disclosure, the support feature contacts the auxiliary heating element in the use position of the auxiliary heating element, such that the support feature generally prevents downward movement of the auxiliary heating element within the cooking cavity.

[0039] According to another aspect of the present disclosure, the auxiliary heating element includes a panel portion and a stop feature that is coupled to the panel portion, wherein contact between the support feature and the stop feature generally prevents forward movement of the auxiliary heating element out of the use position.

[0040] According to another aspect of the present disclosure, the support feature is positioned on an interior side wall of the body and includes a first shelf portion, a second shelf portion between the first shelf portion and an interior rear wall of the body, and a saddle portion extending between the first and second shelf portions.

[0041] According to another aspect of the present disclosure, the stop feature contacts the saddle portion proximate to the first shelf portion in the use position, such that the saddle portion generally prevents forward movement of the auxiliary heating element out of the use position.

[0042] According to yet another aspect of the present disclosure, the panel portion rests upon the first and second shelf portions, such that the first and second shelf portions generally prevent downward movement of the auxiliary heating element within the cooking cavity in the use position of the auxiliary heating element.

[0043] According to another aspect of the present disclosure, a cooking appliance includes a body that defines a cooking cavity, a first electrical connection feature coupled to the body, an auxiliary heating element, and a second electrical connection feature coupled to the auxiliary heating element. The auxiliary heating element is operable between a use position and a removed position. In the use position, the auxiliary heating element is positioned within the cooking cavity. In the removed position, the auxiliary heating element is positioned outside of the cooking cavity. The

second electrical connection feature is operable to move relative to the auxiliary heating element between engaged and disengaged positions and biased toward the disengaged position. Contact between the first electrical connection feature and the second electrical connection feature as the auxiliary heating element enters the use position causes the second electrical connection feature to enter the engaged position. In the use position of the auxiliary heating element, a support feature of the body generally prevents forward movement of the auxiliary heating element, such that the auxiliary heating element is retained in the engaged position.

[0044] According to another aspect of the present disclosure, the second electrical connection feature moves toward the auxiliary heating element from the disengaged position to the engaged position.

[0045] According to another aspect of the present disclosure, the second electrical connection feature includes a pogo pin.

[0046] According to another aspect of the present disclosure, the support feature contacts the auxiliary heating element in the use position of the auxiliary heating element, such that the support feature generally prevents downward movement of the auxiliary heating element within the cooking cavity.

[0047] According to another aspect of the present disclosure, the auxiliary heating element includes a panel portion and a stop feature that is coupled to the panel portion, wherein contact between the support feature and the stop feature generally prevents forward movement of the auxiliary heating element out of the use position.

[0048] According to another aspect of the present disclosure, the support feature is positioned on an interior side wall of the body and includes a first shelf portion, a second shelf portion between the first shelf portion and an interior rear wall of the body, and a saddle portion extending between the first and second shelf portions.

[0049] According to another aspect of the present disclosure, the stop feature contacts the saddle portion proximate to the first shelf portion in the use position, such that the saddle portion generally prevents forward movement of the auxiliary heating element out of the use position.

[0050] According to yet another aspect of the present disclosure, the panel portion rests upon the first and second shelf portions, such that the first and second shelf portions generally prevent downward movement of the auxiliary heating element within the cooking cavity in the use position of the auxiliary heating element.

[0051] According to yet another aspect of the present disclosure, a cooking appliance includes a body that defines a cooking cavity and includes a support feature in communication with the cooking cavity, and an auxiliary heating element operable between a removed position and a use position. In the removed position, the auxiliary heating element is positioned outside of the cooking cavity. In the use position, the auxiliary heating element is positioned within the cooking cavity in contact with the support feature, such that forward movement of the auxiliary heating element out of the use position is generally prevented.

[0052] It will be understood by one having ordinary skill in the art that construction of the described disclosure and other components is not limited to any specific material. Other exemplary embodiments of the disclosure disclosed herein may be formed from a wide variety of materials, unless described otherwise herein.

[0053] For purposes of this disclosure, the term “coupled” (in all of its forms, couple, coupling, coupled, etc.) generally means the joining of two components (electrical or mechanical) directly or indirectly to one another. Such joining may be stationary in nature or movable in nature. Such joining may be achieved with the two components (electrical or mechanical) and any additional intermediate members being integrally formed as a single unitary body with one another or with the two components. Such joining may be permanent in nature or may be removable or releasable in nature unless otherwise stated.

[0054] It is also important to note that the construction and arrangement of the elements of the

disclosure as shown in the exemplary embodiments is illustrative only. Although only a few embodiments of the present innovations have been described in detail in this disclosure, those skilled in the art who review this disclosure will readily appreciate that many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.) without materially departing from the novel teachings and advantages of the subject matter recited. For example, elements shown as integrally formed may be constructed of multiple parts or elements shown as multiple parts may be integrally formed, the operation of the interfaces may be reversed or otherwise varied, the length or width of the structures and/or members or connector or other elements of the system may be varied, the nature or number of adjustment positions provided between the elements may be varied. It should be noted that the elements and/or assemblies of the system may be constructed from any of a wide variety of materials that provide sufficient strength or durability, in any of a wide variety of colors, textures, and combinations. Accordingly, all such modifications are intended to be included within the scope of the present innovations. Other substitutions, modifications, changes, and omissions may be made in the design, operating conditions, and arrangement of the desired and other exemplary embodiments without departing from the spirit of the present innovations.

[0055] It will be understood that any described processes or steps within described processes may be combined with other disclosed processes or steps to form structures within the scope of the present disclosure. The exemplary structures and processes disclosed herein are for illustrative purposes and are not to be construed as limiting.

Claims

1. A cooking appliance, comprising: a body that defines a cooking cavity; a first electrical connection feature coupled to the body; an auxiliary heating element operable between a use position, wherein the auxiliary heating element is positioned within the cooking cavity, and a removed position, wherein the auxiliary heating element is positioned outside of the cooking cavity; and a second electrical connection feature coupled to the auxiliary heating element and operable to move relative to the auxiliary heating element between engaged and disengaged positions, wherein contact between the first electrical connection feature and the second electrical connection feature as the auxiliary heating element enters the use position causes the second electrical connection feature to enter the engaged position.
2. The cooking appliance of claim 1, wherein the second electrical connection feature is biased toward the disengaged position.
3. The cooking appliance of claim 1, wherein the second electrical connection feature moves toward the auxiliary heating element from the disengaged position to the engaged position.
4. The cooking appliance of claim 1, wherein the second electrical connection feature includes a pogo pin.
5. The cooking appliance of claim 1, wherein the body comprises: a support feature in communication with the cooking cavity and configured to support the auxiliary heating element in the use position of the auxiliary heating element.
6. The cooking appliance of claim 5, wherein the support feature contacts the auxiliary heating element in the use position of the auxiliary heating element, such that the support feature generally prevents forward movement of the auxiliary heating element out of the use position.
7. The cooking appliance of claim 6, wherein the support feature contacts the auxiliary heating element in the use position of the auxiliary heating element, such that the support feature generally prevents downward movement of the auxiliary heating element within the cooking cavity.
8. The cooking appliance of claim 7, wherein the auxiliary heating element includes a panel portion and a stop feature that is coupled to the panel portion, wherein contact between the support feature

and the stop feature generally prevents forward movement of the auxiliary heating element out of the use position.

9. The cooking appliance of claim 8, wherein the support feature is positioned on an interior side wall of the body and comprises: a first shelf portion; a second shelf portion between the first shelf portion and an interior rear wall of the body; and a saddle portion extending between the first and second shelf portions.

10. The cooking appliance of claim 9, wherein the stop feature contacts the saddle portion proximate to the first shelf portion in the use position, such that the saddle portion generally prevents forward movement of the auxiliary heating element out of the use position.

11. The cooking appliance of claim 10, wherein the panel portion rests upon the first and second shelf portions, such that the first and second shelf portions generally prevent downward movement of the auxiliary heating element within the cooking cavity in the use position of the auxiliary heating element.

12. A cooking appliance, comprising: a body that defines a cooking cavity; a first electrical connection feature coupled to the body; an auxiliary heating element operable between a use position, wherein the auxiliary heating element is positioned within the cooking cavity, and a removed position, wherein the auxiliary heating element is positioned outside of the cooking cavity; and a second electrical connection feature coupled to the auxiliary heating element and operable to move relative to the auxiliary heating element between engaged and disengaged positions and biased toward the disengaged position, wherein contact between the first electrical connection feature and the second electrical connection feature as the auxiliary heating element enters the use position causes the second electrical connection feature to enter the engaged position, and wherein, in the use position of the auxiliary heating element, a support feature of the body generally prevents forward movement of the auxiliary heating element, such that the auxiliary heating element is retained in the engaged position.

13. The cooking appliance of claim 12, wherein the second electrical connection feature moves toward the auxiliary heating element from the disengaged position to the engaged position.

14. The cooking appliance of claim 12, wherein the second electrical connection feature includes a pogo pin.

15. The cooking appliance of claim 12, wherein the support feature contacts the auxiliary heating element in the use position of the auxiliary heating element, such that the support feature generally prevents downward movement of the auxiliary heating element within the cooking cavity.

16. The cooking appliance of claim 15, wherein the auxiliary heating element includes a panel portion and a stop feature that is coupled to the panel portion, wherein contact between the support feature and the stop feature generally prevents forward movement of the auxiliary heating element out of the use position.

17. The cooking appliance of claim 16, wherein the support feature is positioned on an interior side wall of the body and comprises: a first shelf portion; a second shelf portion between the first shelf portion and an interior rear wall of the body; and a saddle portion extending between the first and second shelf portions.

18. The cooking appliance of claim 17, wherein the stop feature contacts the saddle portion proximate to the first shelf portion in the use position, such that the saddle portion generally prevents forward movement of the auxiliary heating element out of the use position.

19. The cooking appliance of claim 18, wherein the panel portion rests upon the first and second shelf portions, such that the first and second shelf portions generally prevent downward movement of the auxiliary heating element within the cooking cavity in the use position of the auxiliary heating element.

20. A cooking appliance, comprising: a body that defines a cooking cavity and includes a support feature in communication with the cooking cavity; and an auxiliary heating element operable between a removed position, wherein the auxiliary heating element is positioned outside of the

cooking cavity, and a use position, wherein the auxiliary heating element is positioned within the cooking cavity in contact with the support feature, such that forward movement of the auxiliary heating element out of the use position is generally prevented.
